Akhilesh Pothuri

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Experience

Verizon — Remote, United States
Machine Learning Engineer - GenAI — *February 2024 - Present*

- Architected **QAgent** and **QVerse**, enterprise "super agent" harnesses orchestrating specialized sub-agents, secure sandboxes, and persistent memory via **Model Context Protocol (MCP)** and **A2A** communication, effortlessly processing 10,000+ daily queries with sub-second latency.
- Developed a dynamic routing layer using **LangGraph** for complex intent classification, coreference resolution across conversational turns, and progressive prompt disclosure to route queries to the correct specialized sub-agent.
- Engineered a highly-optimized hybrid search pipeline leveraging **pgvector, BM25, & Reciprocal Rank Fusion (RRF k=60)**, integrated with an automated data onboarding pipeline (BigQuery → GCS → Markdown-aware chunking) to feed the knowledge base.
- Built an AI-powered **Data Context Layer** that drastically accelerates database integration by automating SQL schema parsing, semantic enrichment, and documentation mapping with an interactive human-in-the-loop validation interface.
- Led distributed reward-based fine-tuning **(SFT + GRPO)** of Gemma and Llama models on NVIDIA H100 clusters using LoRA adapters. Optimized NLQ-to-SQL workflows with **Chain-of-Thought (CoT)** reasoning, boosting SQL generation accuracy by 35%.
- Built **ThinkForge**, a semantic caching engine utilizing execution-based **VES metrics** to intercept and cache LLM queries, achieving a **60%** cache hit rate and successfully reducing API costs by **40%**.
- Developed an automated **troubleshooting pipeline** utilizing Gemini 3.0 Flash and Whisper to extract structured, actionable diagnostic steps from WebEx recordings via frame detection and audio-visual correlation.
- Productionized FastAPI/Docker architectures on **Kubernetes** and established a rigorous LLM observability framework using **RAGAS** to monitor faithfulness, latency, and hallucination rates in real-time.

UNC Charlotte Football — Charlotte, NC
Data Scientist — *August 2023 - December 2023*

- Built an automated live tracking system using **Convolutional Neural Networks (CNN)** & advanced image processing algorithms (**computer vision**) for real-time player & jersey number recognition, saving 5 hours daily.
- Employed advanced statistical models, including **regression** and **clustering** to assess player strengths, weaknesses, and overall contribution to the team's performance.
- Utilized **SQL** for complex data manipulation tasks, including merging, aggregation, & **KPI** calculation for players.

Hewlett Packard R&D — Bangalore, India
Data Scientist — *February 2020 - July 2022*

- Led POC project to analyze global call volume data using **time series forecasting**, achieving a 28% mape boost.
- Conducted **operational analysis** to enhance risk mitigation & resource allocation, leveraging data-driven insights for informed decision-making. Employed predictive analytics for strategic optimization of supply chain resources.
- Performed ad-hoc data analysis in Python, fabricated a **data warehouse** & optimised SQL queries in **BigQuery**
- Developed and maintained financial models for forecasting, **budget planning**, and scenario analysis, leading to 15% improvement in budget accuracy and 20% reduction in forecasting errors.
- Built a spare part replacement **recommendation engine** with **natural language processing (NLP)** such as Transformers, reducing customer wait time to 15s enhancing user experience.
- Fine-tuned a **PyTorch based LSTM** model within the recommendation engine, optimizing its performance in understanding sequential patterns and improving the accuracy of personalized recommendations.
- Applied Dimensionality reduction & One-Class SVM for **anomaly detection** in complex, high-dimensional data.
- Spearheaded **end-to-end** development and deployment of **Demand Forecasting** projects, employing advanced techniques like ABC/XYZ classification and ADI/CV square-based classification for precise analysis.
- Executed comprehensive analysis on demand forecasting data, leveraging statistics and machine learning techniques to optimize **inventory management**.
- Employed monthly snapshots of spare parts demand for exploratory data analysis (EDA), **feature engineering** & modeling in Python & R, yielding an anticipated **$26M** ROI over 3 years, optimizing supply chain efficiency.
- Automated **ETL** processes, model training, & evaluation via scalable **end-to-end** data infrastructure pipelines in **Azure Databricks** & **Azure Data Factory**, leveraging **Spark** clusters, saving 120 labor hours monthly.
- Delivered **actionable insights** to cross-functional teams, stakeholders through interactive **PowerBI** dashboards

Skills

- **Agentic AI & Orchestration:** ReAct, Chain-of-Thought (CoT), Planner-Orchestrator-Reflector patterns; LangGraph, LangChain, Model Context Protocol (MCP); Intent Routing, Multi-agent Task Decomposition, and Autonomous Workflows.
- **LLM Fine-Tuning & Training:** Supervised Fine-Tuning (SFT), PEFT (LoRA, QLoRA), Reinforcement Learning from Human Feedback (RLHF); Proximal Policy Optimization (PPO, GRPO); Distributed training on NVIDIA H100 clusters.
- **RAG & Retrieval Systems:** Retrieval-Augmented Generation (RAG), Semantic & Hybrid Search, pgvector, ChromaDB, FAISS; Layout-aware parsing, Metadata filtering, and Citation grounding.
- **LLM Evaluation & Observability:** Evaluation Frameworks (RAGAS, automated benchmarking suites); Metrics for Accuracy, Relevance, Hallucination Detection, and Latency; Systematic A/B Testing.
- **Machine Learning & Data Science:** Deep Learning (CNN, LSTM), Computer Vision (OpenCV), Time Series Forecasting; Predictive Modeling, Clustering, and Anomaly Detection.
- **Programming & Frameworks:** Python (Pandas, PyTorch, TensorFlow, Scikit-Learn, FastAPI), R, SQL; Async programming and RESTful API design.
- **Data Infrastructure & Cloud:** Azure (Databricks, Data Factory), GCP (BigQuery, Cloud Storage), AWS S3; PostgreSQL, Snowflake, Oracle, Teradata; Docker, Kubernetes, GitLab CI/CD.

Projects

Nexus-Graph [demo]

- Developed an **open-source CLI tool and MCP server** that optimizes AI context retrieval, reducing token usage by up to **95%** through a surgical graph-based mapping system.
- Implemented **Abstract Syntax Tree (AST)** parsing using Tree-sitter to map code symbols (functions, classes, dependencies) into a high-performance SQLite-backed relational graph.
- Distributed via **NPM** and built a high-fidelity marketing site with an interactive Vis.js simulation dashboard to visualize complex code dependencies.

CostLine -AI Spend Intelligence & FinOps Platform [demo]

- Built an infrastructure-level proxy and intelligence platform that enforces hard budget limits (sub-10ms overhead) and tracks per-user/per-feature unit economics for LLM applications.
- Implemented automated optimization detection, analyzing traffic for RAG over-retrieval, history bloat, and model routing inefficiencies to provide actionable cost-saving recommendations.
- Designed the platform for enterprise self-hosting (Docker, Kubernetes, Terraform for AWS/GCP/Azure) to ensure zero customer data exfiltration for air-gapped environments.

localGPT (GitHub Open-Source Contribution)

- Developed and integrated a custom **JSON loader class** to streamline document ingestion and automated embedding generation using **Chroma vectorDB**.

Awards

- Third Prize Winner, Reinvent-The-Wheel 2.0 Hackathon (24-hour sprint) hosted by TORQATA & ATD; competed against 20+ professional data science teams. (Nov 2022)
- **Akhilesh Pothuri** and Pooja Malik. Knowledge graph augmented advanced learning models for commonsense reasoning. In *IOP Conference Series: Materials Science and Engineering*, volume 1022, page 012038. IOP Publishing, 2021[**Paper**]

Education

University of North Carolina at Charlotte — Charlotte, North Carolina
Master of Science in Data Science and Business Analytics — *August 2022 - December 2023*

Shiv Nadar University — Greater Noida, India
Bachelor of Technology in Computer Science and Engineering, Minor in Mathematics — *July 2016 - July 2020*